

Regularization, Generalization, and Trustworthy Evaluation

Week 5: Evidence-Based Reliability

From Training Success To Trustworthy Evaluation

This section covers these main ideas.

- ▶ Motivation And Prior Knowledge Activation

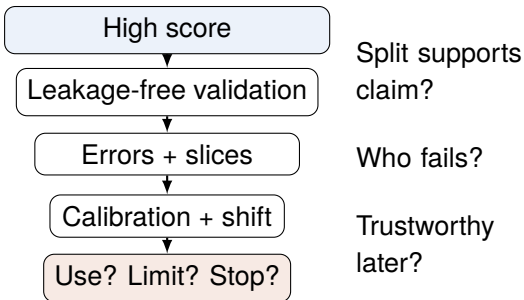
From Training Success To Trustworthy Evaluation

Focus: Motivation And Prior Knowledge Activation

Motivation And Prior Knowledge Activation

Week 5 Question – Can We Trust This Model?

- Last week: task, data, stakeholders, risks, assumptions.
- This week: test whether behavior supports those assumptions.
- A strong score can hide leakage, subgroup failure, overconfidence, or shift risk.

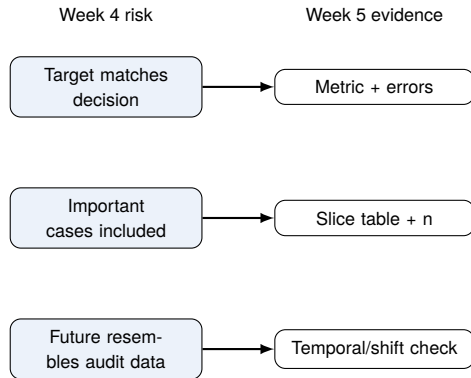


Takeaway.

Discussion: why can 92% validation accuracy still be unacceptable for deployment?

Evaluation Evidence Links Back To The Audit

- Charter assumption: does prediction target match a real decision?
- Audit concern: are important cases missing or biased?
- Evidence stream now includes: curves, metrics, errors, slices, calibration, shift checks.
- The portfolio audit is not a static document; it is updated by evidence.



Review: Generalization Failure Modes

This section covers these main ideas.

- ▶ Review: Generalization And Learning Curves
- ▶ Review: Overfitting And Underfitting

Review: Generalization Failure Modes

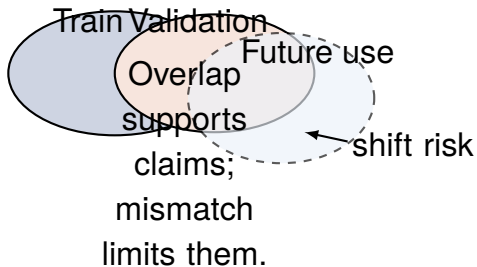
Focus: Review: Generalization And Learning Curves

Review: Generalization And Learning Curves

- Review: Overfitting And Underfitting

Review: From Curves To Claims

- You have already seen training and validation curves as debugging evidence.
- Today we reuse them to decide what reliability claim is defensible.
- Validation estimates future behavior only under split assumptions.
- Deployment can differ by time, population, policy, or measurement.



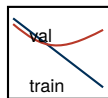
Takeaway.

Review point: validation is an assumption test, not a deployment guarantee.

Review: Curve Patterns

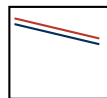
- Divergence asks for split audit, regularization, early stopping, or simplification evidence.
- Plateau asks whether features, labels, capacity, or optimization limit the task.
- Instability asks for sample-size, logging, and update-stability checks.
- Week 5 move: name the pattern, test one response, and state the limit.

Divergence



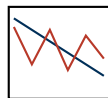
epoch

Plateau



epoch

Instability



epoch

Review: Generalization Failure Modes

Focus: Review: Overfitting And Underfitting

- Review: Generalization And Learning Curves

Review: Overfitting And Underfitting

Review: Failure Mode To Response

Pattern	Reliability risk	Next evidence	Possible response
Train improves, val worsens Both remain poor	Memorization or shortcut learning Task, features, labels, or capacity may limit the claim	Split audit, errors, duplicate or leakage check Error examples, feature audit, baseline comparison	Regularize, stop early, simplify, clean data Improve inputs, adjust model, narrow scope
Validation noisy	Estimate may be unstable	Sample size, class balance, repeated-split evidence	Report uncertainty, collect data, simplify claim
High score, risky slices	Aggregate hides deployment failure	Slice table with sample sizes and examples	Limit deployment, redesign, gather evidence

Takeaway.

Review is only useful if it becomes an evaluation action.

Regularization As Controlled Constraint

This section covers these main ideas.

- ▶ Regularization Principle
- ▶ Implementation Patterns

Regularization As Controlled Constraint

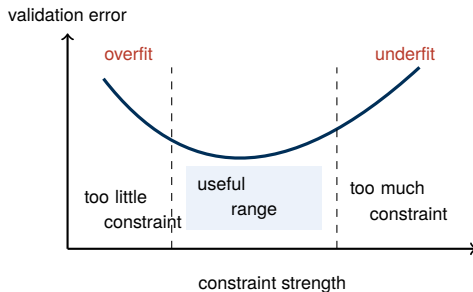
Focus: Regularization Principle

Regularization Principle

- Implementation Patterns

Regularization Trades Fit For Robustness

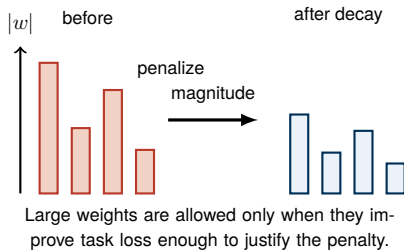
- Add just enough constraint to reduce memorization.
- Optimize held-out behavior, not only training fit.
- Too little constraint can overfit.
- Too much constraint can underfit.
- Compare against the same validation design.



Weight Decay Penalizes Large Weights

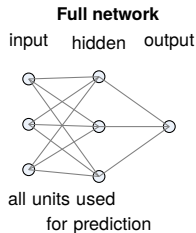
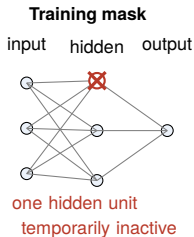
- Weight decay adds a penalty for large parameters during training.
- Smoother parameter settings can improve generalization.
- Useful for high-capacity models that fit noise.
- It does not replace fixing leakage, wrong splits, or bad labels.

$$L_{total} = L_{task} + \lambda \|w\|_2^2$$

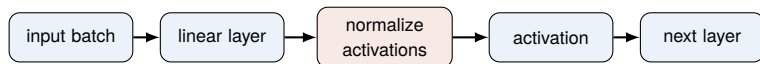


Dropout Reduces Co-Adaptation

- Training: randomly disable units.
- This prevents hidden units from depending too much on one partner unit.
- Evaluation: use the learned ensemble-like behavior, not random deletion.
- Best fit: high-capacity models with co-adaptation risk.
- Does not fix leakage or distribution shift.



Normalization Layers Stabilize Activations



Normalization makes activation scale less surprising before the next transformation.

- BatchNorm normalizes layer activations using mini-batch statistics during training.
- LayerNorm normalizes within each example and is common in sequence models and transformers.
- Main role: smoother, less scale-sensitive training.
- Regularizing side effects can help, but they do not replace leakage-free evaluation.

Normalization: What Changes?

$$\hat{z} = \frac{z - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

$$\tilde{z} = \gamma \hat{z} + \beta, \quad a = \sigma(\tilde{z})$$

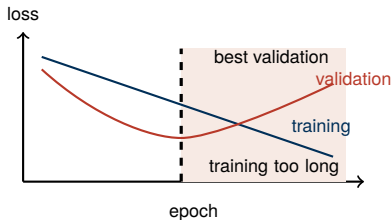
- Center and rescale pre-activations before the nonlinearity.
- Learn γ and β so the layer can recover useful pre-activation scale.
- Regularizing effect is indirect, not a substitute for validation design.

Variable map

z	pre-activation from a linear layer
$\hat{z}$	normalized pre-activation
$\tilde{z}$	shifted and scaled pre-activation
a	activation after σ
γ, β	learned scale and shift
μ, σ^2	statistics from the normalized axis

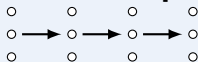
Early Stopping Stops Before Memorization

- Track validation across epochs.
- Restore the best validation epoch.
- Stopping is a constraint on memorization.
- Do not repeatedly tune against the test set.



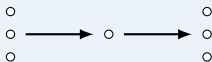
Model Shape Controls Complexity

Width + depth



Layer/unit counts
change complexity.

Bottleneck



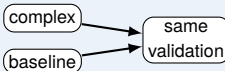
Narrow middle forces
compact representation.

Feature path



Remove noisy shortcut features.

Baseline



Small gain? Keep baseline.

Model Shape Decision Details

Shape choice	Examples	Can help when	Reliability caution
Width/depth	Fewer layers or units	Overfitting or small data	May underfit if too small
Bottleneck	Narrow hidden layer	Need compact representation	Can discard signal
Feature path	Skip, embedding, interaction limits	Shortcut interactions	Compare on same split
Baseline	Linear or shallow model	Similar validation evidence	Prefer if gain is small

Takeaway.

Model shape is regularization: choose complexity by evidence and risk.

Regularization As Controlled Constraint

Focus: Implementation Patterns

- ▶ Regularization Principle

- Implementation Patterns**

PyTorch: Regularization Knobs

```
model = nn.Sequential(  
    nn.Linear(n_features, 64),  
    nn.ReLU(),  
    # p=0.25 means each activation has a 25% drop chance.  
    # Dropout usually comes after the activation.  
    nn.Dropout(p=0.25),  
    nn.Linear(64, n_classes),  
)  
  
optimizer = torch.optim.AdamW(  
    # weight_decay is the optimizer's lambda-like penalty strength.  
    model.parameters(), lr=1e-3, weight_decay=1e-4  
)
```

- `weight_decay` is the practical knob corresponding to λ in the penalty slide.
- `Dropout (p=0.25)` drops about 25% of activations during training.

PyTorch: Shape Choices

```
model = nn.Sequential(  
    nn.Linear(n_features, 32),      # width choice  
    nn.ReLU(),  
    nn.Linear(32, 16),              # depth + bottleneck choice  
    nn.ReLU(),  
    nn.Linear(16, n_classes),  
)
```

- Width: number of units in a layer.
- Depth: number of learned layers.
- Bottleneck: a deliberately narrow middle representation.

PyTorch: Normalization Layers

```
model = nn.Sequential(  
    nn.Linear(n_features, 32),  
    nn.LayerNorm(32),          # normalize within each example  
    nn.ReLU(),  
    nn.Linear(32, n_classes),  
)  
  
# For tabular mini-batches, BatchNorm1d(32) is another option.
```

Takeaway.

Normalization changes the activation scale; validation decides whether it helped.

PyTorch: Early Stopping Pattern

```
# PyTorch has no built-in early-stopping switch.  
# You implement patience in the training loop or use a helper library.  
if val_loss < best_val_loss:  
    best_state = copy.deepcopy(model.state_dict())  
    best_val_loss = val_loss  
    epochs_without_improvement = 0  
else:  
    epochs_without_improvement += 1  
  
if epochs_without_improvement >= patience:  
    break
```

Takeaway.

Stopping is chosen from validation behavior, not from the test set.

Validation Design And Leakage Control

This section covers these main ideas.

- ▶ Split Strategy As A Claim

Validation Design And Leakage Control

Focus: Split Strategy As A Claim

Split Strategy As A Claim

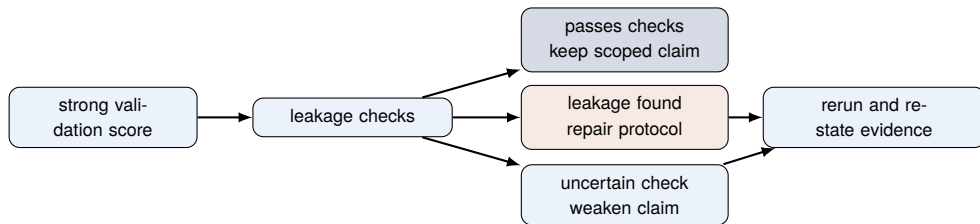
Split Assumptions

Split	Use when	Claim it supports
Random	Cases are independent and exchangeable	Future resembles shuffled history
Cross-val	Data is small and variance matters	Estimate is less split-dependent
Temporal	Past predicts future cases	Later cases are a real test
Grouped	Users, patients, devices, or documents repeat	Entities do not leak across splits

Takeaway.

Assignment 5 should explain why the split matches the intended future use.

Leakage Checks Are Claim Gates

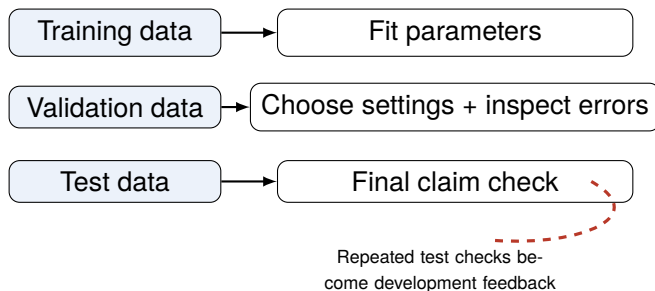


Week 5 addition: leakage checks decide whether the evaluation result supports a reliability claim, only an exploratory claim, or no claim yet.

Takeaway.

A lower score after repair can be more trustworthy than a high leaky score.

Validation Guides, Test Checks



Takeaway.

Assignment 5 should state which evidence came from validation and what came from a held-out test.

Validation Results Need Uncertainty And Context

Report	Accuracy	Sample context	Responsible claim
A	0.91	1,200 cases, balanced classes	Stronger estimate, still scoped to split
B	0.91	48 cases, rare positives	Fragile estimate; needs uncertainty/context

Takeaway.

A metric is evidence for a claim; it is not the whole claim.

Error, Subgroup, Slice, And Calibration Analysis

This section covers these main ideas.

- ▶ From Aggregate Metrics To Failure Patterns

Error, Subgroup, Slice, And Calibration Analysis

Focus: From Aggregate Metrics To Failure Patterns

From Aggregate Metrics To Failure Patterns

Error Analysis Turns Mistakes Into Evidence

Example	Pred.	True	Error type	Suspected cause	Action
Case A	approve	deny	stakeholder-critical	missing risk signal	add review rule
Case B	deny	approve	boundary	near threshold	inspect threshold
Case C	class 2	class 1	data quality	inconsistent label	audit labels

Takeaway.

Examples should create testable failure hypotheses, not replace metrics.

Slice Checks Reveal Uneven Performance

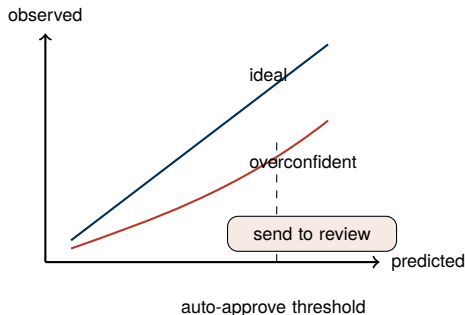
Group or slice	n	Accuracy	Responsible reading
Aggregate	900	0.89	Looks acceptable overall
Returning users	520	0.92	Supported by enough evidence
New users	80	0.71	Concentrated failure; investigate
Rare product type	12	0.58	Too small for hard claim; flag risk

Takeaway.

Do not invent demographic or fairness claims when metadata is missing.

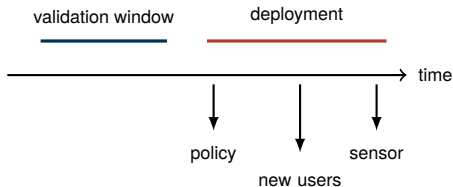
Calibration Guides Action Thresholds

- If 90% confidence is only 70% correct, automation is too risky.
- Use calibrated scores to set auto-approve, human review, or reject bands.
- Poor calibration means recalibrate, raise the review threshold, or avoid probability claims.



Distribution Shift Makes Reliability Conditional

- Future inputs, labels, policies, or behavior may change.
- Time, location, population, sensor, and process shifts are common.
- Reliability claims need boundaries and monitoring triggers.



Reliability Judgment And Assignment 5 Alignment

This section covers these main ideas.

- ▶ From Evidence To Recommendation

Reliability Judgment And Assignment 5 Alignment

Focus: From Evidence To Recommendation

From Evidence To Recommendation

Reliability Judgment Is Conditional

Claim level	Supports use	Argues against use	Still unknown
Week 4 fix claim	Controlled before/after trace	One failure only	Other risks
Week 5 reliability claim	Leakage-free split, stable curves	Severe errors, poor calibration	Missing metadata
Deployment claim	Acceptable slices, audit alignment	Weak sample size, shift risk	Monitoring gap

Takeaway.

Recommendation: deploy, pilot, gather more evidence, redesign, or do not deploy.

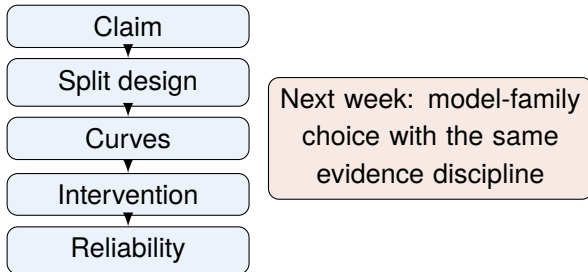
Assignment 5 Walkthrough

- ☐ Diagnose curves and validation behavior
- ☐ Justify one intervention
- ☐ Explain split roles and leakage controls
- ☐ Analyze errors plus subgroup or slice
- ☐ Write reliability judgment and audit update

Takeaway.

If metadata is missing, report the limitation instead of claiming subgroup reliability.

Recap And Next Week Preview



Takeaway.

Exit ticket: what result would make you trust the model less despite a good aggregate score?